Frozen office-validation-v6: complete traces, 10 ms peak envelopes
Contact is summed force magnitude; failure tails retained. Shading = commanded hand / rod intervals.

Tracking position error

mm

Maximum absolute joint actuator torque

Nm

End-effector linear speed

m/s

Human / robot contact load

N

Rod / robot contact load

N

Unknown-edge contact load

N

Time (s)

Time (s)

ESN seed 1 (failed)
ESN seed 2 (task only)
Matched RNN seed 1 (failed)
Matched RNN seed 2 (strict)